

# Preparation Notes for “IP8: Metagenomic sequence analysis using *k*-mer based methods.”

Some sessions and tools require no preparation. This document covers the setup, datasets, and environment configuration needed only for those that do. For questions, email me (Ali Şapcı) at [asapci\(at\)ucsd\(dot\)edu](mailto:asapci@ucsd.edu) or open an issue in this repository.

**System requirements:** Participants should have access to a Linux or macOS system with at least 16 GB of RAM and 16 GB of free disk space to run the tutorial.

All participants must have conda/mamba or miniconda/micromamba installed. See the [miniconda installation guide](#) (or the [micromamba installation guide](#)) if you need to set this up. Before the tutorial, we suggest that participants install sourmash, YACHT (**Part 3**), sylph, skani (**Part 4**), krepp (**Part 5**), and kf2vec (**Part 7**) as described below.

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**Part 1 - Kraken2 (Ben Langmead):** No preparation is needed; the tutorial will be on an interactive page directly using a web browser.

**Part 2 - VirFinder and d2 (Fengzhu Sun):** No preparation or pre-installation is needed. The tutorial will be conducted through [this Colab notebook](#). To access seamlessly, log in to a Google account to use the interactive notebook.

Optional: Participants may review [this protocol paper](#): Mo, Y., Ahlgren, N., Fuhrman, J. A., Sun, F., & Hou, S. (2026). A beginner's guide to using DeepVirFinder for viral sequence identification from metagenomic datasets. *Current Protocols*, 6, e70310. doi: [10.1002/cpz1.70310](https://doi.org/10.1002/cpz1.70310). Please email me at [asapci\(at\)ucsd\(dot\)edu](mailto:asapci@ucsd.edu) if you do not have access (e.g., due to a paywall).

**Part 3 - sourmash and YACHT - (David Koslicki):** Please follow the steps given [in this repository](#) to install the software and download the dataset for the exercises.

**Part 4 - sylph and skani (Yun William Yu):** Install sylph and skani following the instructions [on this page](#) and using the command [here](#), respectively.

**Part 5 - krepp (Ali Osman Berk Şapcı):** Please follow the steps [on this page](#) to install the software and download the dataset for the exercises.

**Part 6 - DecoDiPhy (Shayesteh Arasti):** No preparation or pre-installation is needed. The DecoDiPhy tutorial will follow this [Colab notebook](#). Make sure you have a Google account and can use Google Colab.

**Part 7 - kf2vec (Eleonora Rachtman):** Follow the steps on [this page](#) to install `kf2vec` using `conda` and clone the GitHub repository for the toy example.